

A Complete Polynomial Certificate for Rank-One Scaling of Determinantal Block Tensors

1 Statement of the problem and main theorem

Fix an integer $n \geq 5$. Let $A^{(1)}, \dots, A^{(n)} \in \mathbb{R}^{3 \times 4}$ be *Zariski-generic* (in particular, every 3×4 matrix has rank 3 and all genericity statements below hold).

For $\alpha \in [n] := \{1, \dots, n\}$ and $i \in \{1, 2, 3\}$ write

$$a_{(\alpha, i)} := A^{(\alpha)}(i, :) \in \mathbb{R}^4.$$

For $\alpha, \beta, \gamma, \delta \in [n]$ and $i, j, k, \ell \in \{1, 2, 3\}$ define the $3 \times 3 \times 3 \times 3$ *determinantal block* tensor

$$Q_{ijkl}^{(\alpha\beta\gamma\delta)} := \det(a_{(\alpha, i)}, a_{(\beta, j)}, a_{(\gamma, k)}, a_{(\delta, \ell)}).$$

Let $\lambda \in \mathbb{R}^{n \times n \times n \times n}$ satisfy the *support condition*

$$\lambda_{\alpha\beta\gamma\delta} \neq 0 \quad \text{iff} \quad (\alpha, \beta, \gamma, \delta) \text{ are not all equal,}$$

and define observed blocks

$$T_{ijkl}^{(\alpha\beta\gamma\delta)} := \lambda_{\alpha\beta\gamma\delta} Q_{ijkl}^{(\alpha\beta\gamma\delta)}.$$

1.1 Global tensor encoding

Let $V := \mathbb{R}^{3n}$ with standard basis vectors $e_{(\alpha, i)}$ indexed by $(\alpha, i) \in [n] \times \{1, 2, 3\}$. Define the *global tensor* $\mathbf{T} \in V^{\otimes 4}$ by

$$\mathbf{T}_{(\alpha, i), (\beta, j), (\gamma, k), (\delta, \ell)} := T_{ijkl}^{(\alpha\beta\gamma\delta)}.$$

1.2 Goal

We wish to decide whether λ has rank 1 as a 4-way tensor, i.e. whether there exist vectors $u, v, w, x \in (\mathbb{R}^*)^n$ such that

$$\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta \quad \text{for all } (\alpha, \beta, \gamma, \delta) \text{ not all equal.} \tag{1}$$

1.3 Universal polynomial test

We will define an explicit polynomial map F in the entries of $\mathbf{T}$ (hence in the observed data), with bounded degree independent of n , such that $F(\mathbf{T}) = 0$ holds *if and only if* (1) holds for Zariski-generic $A^{(\alpha)}$.

Definition 1 (Mode flattenings). For $m \in \{1, 2, 3, 4\}$, the *mode- m flattening* of $\mathbf{T}$ is the matrix

$$\text{Flat}_m(\mathbf{T}) \in \mathbb{R}^{(3n) \times (3n)^3}$$

obtained by viewing $\mathbf{T}$ as a linear map $V \rightarrow V^{\otimes 3}$ in the m -th slot.

Definition 2 (Two-mode slices). Fix an ordered pair of distinct modes (p, q) with complement modes (r, s) . For fixed indices $i_r \in [3n]$ in mode r and $i_s \in [3n]$ in mode s , define the (p, q) -slice matrix

$$M_{(p,q)}^{(i_r, i_s)} \in \mathbb{R}^{(3n) \times (3n)}$$

by

$$(M_{(p,q)}^{(i_r, i_s)})_{a,b} := \mathbf{T}_{\dots, a, \dots, b, \dots, i_r, \dots, i_s, \dots}$$

where a occupies mode p , b occupies mode q , and i_r, i_s occupy modes r, s .

Definition 3 (Universal polynomial map F). Let $F(\mathbf{T})$ be the concatenation of the following polynomial families:

(F1) Flattening minors (degree 5). For each $m \in \{1, 2, 3, 4\}$, include all 5×5 minors of $\text{Flat}_m(\mathbf{T})$.

(F2) Slice minors (degree 3). For each ordered pairing (p, q) with complement (r, s) , and each fixed $(i_r, i_s) \in [3n]^2$, include all 3×3 minors of the slice matrix $M_{(p,q)}^{(i_r, i_s)}$.

Thus $F(\mathbf{T}) = 0$ means simultaneously:

$$\text{rank}(\text{Flat}_m(\mathbf{T})) \leq 4 \quad \forall m \quad \text{and} \quad \text{rank}(M_{(p,q)}^{(i_r, i_s)}) \leq 2 \quad \forall (p, q), (i_r, i_s).$$

Theorem 1 (Main theorem). Fix $n \geq 5$ and let $A^{(1)}, \dots, A^{(n)} \in \mathbb{R}^{3 \times 4}$ be Zariski-generic. Let λ satisfy the support condition (nonzero off the full diagonal) and define $\mathbf{T}$ as above. Then:

(a) If λ satisfies (1), then $F(\mathbf{T}) = 0$.

(b) Conversely, if $F(\mathbf{T}) = 0$, then λ satisfies (1).

In particular, for generic $A^{(\alpha)}$, the condition $F(\mathbf{T}) = 0$ is a universal bounded-degree polynomial certificate for rank-1 of λ (off the full diagonal).

The remainder of the document proves Theorem 1 with explicit intermediate lemmas and no omitted justifications, except for explicitly labeled standard linear-algebra facts and the fundamental theorem of projective geometry (which we also prove in the special form needed).

2 Necessity: rank-one λ implies $F(\mathbf{T}) = 0$

Assume (1) holds. Define linear maps $B_m : V \rightarrow \mathbb{R}^4$ by

$$B_1(e_{(\alpha, i)}) = u_\alpha a_{(\alpha, i)}, \quad B_2(e_{(\beta, j)}) = v_\beta a_{(\beta, j)}, \quad B_3(e_{(\gamma, k)}) = w_\gamma a_{(\gamma, k)}, \quad B_4(e_{(\delta, \ell)}) = x_\delta a_{(\delta, \ell)}. \quad (2)$$

Then for all indices,

$$\det(B_1 e_{(\alpha, i)}, B_2 e_{(\beta, j)}, B_3 e_{(\gamma, k)}, B_4 e_{(\delta, \ell)}) = u_\alpha v_\beta w_\gamma x_\delta \det(a_{(\alpha, i)}, a_{(\beta, j)}, a_{(\gamma, k)}, a_{(\delta, \ell)}),$$

which equals $\mathbf{T}_{(\alpha, i), (\beta, j), (\gamma, k), (\delta, \ell)}$ by definition. Hence:

Lemma 1 (Determinant pullback representation). If (1) holds, then

$$\mathbf{T}_{abcd} = \det(B_1 e_a, B_2 e_b, B_3 e_c, B_4 e_d) \quad \forall a, b, c, d \in [3n]. \quad (3)$$

2.1 Flattening rank bound

Lemma 2 (Flattening rank ≤ 4). *Under (3), for each mode m , $\text{rank}(\text{Flat}_m(\mathbf{T})) \leq 4$.*

Proof. Fix $m = 1$ (others identical). View $\mathbf{T}$ as a linear map $V \rightarrow V^{\otimes 3}$: for $u \in V$ and basis elements $b, c, d \in [3n]$,

$$(\text{Flat}_1(\mathbf{T})u)_{bcd} = \mathbf{T}(u, e_b, e_c, e_d) = \det(B_1u, B_2e_b, B_3e_c, B_4e_d).$$

For fixed (b, c, d) , the map $u \mapsto \det(B_1u, \cdot, \cdot, \cdot)$ depends on u only through $B_1u \in \mathbb{R}^4$, and is linear in B_1u . Thus $\text{Flat}_1(\mathbf{T})$ factors as

$$V \xrightarrow{B_1} \mathbb{R}^4 \xrightarrow{L} V^{\otimes 3}$$

for some linear L , hence $\text{rank}(\text{Flat}_1(\mathbf{T})) \leq \text{rank}(B_1) \leq 4$. Therefore all 5×5 minors vanish. $\square$

2.2 Two-mode slice rank bound

Lemma 3 (Rank ≤ 2 for every two-mode slice). *Under (3), for every ordered pairing (p, q) and every fixed indices in the other two modes, the corresponding slice matrix has rank ≤ 2 .*

Proof. It suffices to prove for $(p, q) = (1, 2)$ with fixed (c, d) in modes $(3, 4)$. Let $z_3 := B_3e_c$, $z_4 := B_4e_d \in \mathbb{R}^4$. Define the bilinear form on $\mathbb{R}^4 \times \mathbb{R}^4$:

$$\beta_{z_3, z_4}(x, y) := \det(x, y, z_3, z_4).$$

Choose a basis of $\mathbb{R}^4$ so that z_3, z_4 span the last two coordinate directions. Then $\beta_{z_3, z_4}(x, y)$ depends only on the first two coordinates of x and y , hence has matrix rank ≤ 2 as an element of $\mathbb{R}^4 \otimes \mathbb{R}^4$. Equivalently, the linear map $x \mapsto \beta_{z_3, z_4}(x, \cdot) \in (\mathbb{R}^4)^*$ has rank ≤ 2 . Now the slice matrix is

$$M_{ab}^{(c, d)} = \mathbf{T}_{abcd} = \det(B_1e_a, B_2e_b, z_3, z_4) = \beta_{z_3, z_4}(B_1e_a, B_2e_b).$$

Thus $M^{(c, d)} = B_1^\top K(z_3, z_4) B_2$ where $K(z_3, z_4)$ is a 4×4 matrix of rank ≤ 2 representing β_{z_3, z_4} . Therefore $\text{rank}(M^{(c, d)}) \leq \text{rank}(K(z_3, z_4)) \leq 2$. The same argument applies to all pairings. Hence all 3×3 minors vanish. $\square$

Combining Lemmas 2 and 3 yields $F(\mathbf{T}) = 0$, proving Theorem 1(a).

3 Sufficiency roadmap

Assume now $F(\mathbf{T}) = 0$. We must prove (1). The proof proceeds in four major steps:

Step 1: (§4) From flattening minors (F1), obtain Tucker-through-4:

$$\mathbf{T} = G \times_1 B_1 \times_2 B_2 \times_3 B_3 \times_4 B_4 \quad \text{with} \quad B_m : V \rightarrow \mathbb{R}^4, \quad G \in (\mathbb{R}^4)^{\otimes 4}.$$

Step 2: (§5) From slice minors (F2'), show that *every* 4×4 slice of the core G in every pairing has rank ≤ 2 .

Step 3: (§6) Prove the *core rigidity theorem*: such a G (with full multilinear rank 4) is $GL(4)^4$ -equivalent to the Levi-Civita determinant core ε (possibly after a global dualization). This yields a determinant pullback representation of $\mathbf{T}$.

Step 4: (§7) Use the block structure (α, i) and genericity of $A^{(\alpha)}$ to force each B_m to be block-scaled, hence λ is rank-one.

4 Tucker reduction from flattening minors

We use a standard multilinear algebra fact: a tensor has multilinear rank bounded by (r_1, r_2, r_3, r_4) iff each mode- m flattening has matrix rank $\leq r_m$, and then a Tucker decomposition exists.

Lemma 4 (Tucker decomposition from flattening ranks). *Let $\mathbf{T} \in V^{\otimes 4}$ with $V = \mathbb{R}^{3n}$. Suppose $\text{rank}(\text{Flat}_m(\mathbf{T})) \leq 4$ for all m . Then there exist linear maps $B_m : V \rightarrow \mathbb{R}^4$ and a core $G \in (\mathbb{R}^4)^{\otimes 4}$ such that*

$$\mathbf{T} = G \times_1 B_1 \times_2 B_2 \times_3 B_3 \times_4 B_4. \quad (4)$$

Moreover, on a Zariski-open set (the generic case), each B_m has rank 4.

Proof. For each m , let $U_m \subset V$ be the column space of $\text{Flat}_m(\mathbf{T})$; then $\dim U_m \leq 4$. Choose a linear isomorphism $\phi_m : U_m \rightarrow \mathbb{R}^4$ (extend by adding a complementary subspace if $\dim U_m < 4$). Let $B_m := \phi_m \circ \pi_{U_m}$ where $\pi_{U_m} : V \rightarrow U_m$ is a linear projection. Then B_m has rank ≤ 4 . Define G by restricting $\mathbf{T}$ to $U_1 \times U_2 \times U_3 \times U_4$ and transporting via ϕ_m . Explicitly,

$$G(x_1, x_2, x_3, x_4) := \mathbf{T}(\phi_1^{-1}x_1, \phi_2^{-1}x_2, \phi_3^{-1}x_3, \phi_4^{-1}x_4).$$

Then by construction $\mathbf{T}$ equals the Tucker reconstruction (4). Generic rank-4 holds on a Zariski-open set where each flattening rank equals 4. $\square$

Since (F1) is exactly the vanishing of all 5×5 flattening minors, Lemma 4 applies.

5 Slice-rank constraints descend to the core

Assume $\mathbf{T}$ admits a Tucker form (4) with $\text{rank}(B_m) = 4$ (generic case). Fix a pairing (p, q) with complement (r, s) and fixed indices in modes (r, s) . The corresponding slice matrix is a conjugation of the corresponding core slice matrix by B_p, B_q .

Lemma 5 (Slices conjugate to core slices). *Assume (4) and $\text{rank}(B_p) = \text{rank}(B_q) = 4$. Fix indices (i_r, i_s) in modes (r, s) . Then there exists a 4×4 core slice matrix $G_{(p,q)}^{(i_r, i_s)}$ such that*

$$M_{(p,q)}^{(i_r, i_s)} = B_p^\top G_{(p,q)}^{(i_r, i_s)} B_q.$$

Proof. Write the contraction of $\mathbf{T}$ in modes (r, s) against basis vectors e_{i_r}, e_{i_s} to obtain a bilinear form in modes (p, q) . Under (4), contraction commutes with mode multiplications and yields exactly the core slice contracted against $B_r e_{i_r}$ and $B_s e_{i_s}$; the resulting matrix in the latent coordinates is $G_{(p,q)}^{(i_r, i_s)}$. Then expanding in the ambient bases gives $B_p^\top(\cdot)B_q$ as claimed. $\square$

Lemma 6 (Slice-rank constraints descend). *Assume (4) and $\text{rank}(B_m) = 4$ for all m . If every ambient slice matrix $M_{(p,q)}^{(i_r, i_s)}$ has rank ≤ 2 (i.e. all its 3×3 minors vanish), then every corresponding core slice matrix $G_{(p,q)}^{(i_r, i_s)}$ has rank ≤ 2 .*

Proof. By Lemma 5, $M = B_p^\top G B_q$ for some 4×4 matrix G and full column-rank B_p, B_q . Choose left inverses L_p, L_q on the images: $L_p^\top B_p^\top = \text{Id}_4$ and $B_q L_q = \text{Id}_4$ (possible since rank 4). Then

$$G = L_p^\top M L_q.$$

Therefore $\text{rank}(G) \leq \text{rank}(M) \leq 2$. $\square$

Thus: under $F(\mathbf{T}) = 0$ and generic rank 4 of B_m , every 4×4 core slice matrix in every pairing has rank ≤ 2 .

6 Core rigidity via explicit Grassmannian line geometry

This section proves:

Theorem 2 (Core rigidity). *Let V_1, V_2, V_3, V_4 be 4-dimensional real vector spaces and let $G \in V_1 \otimes V_2 \otimes V_3 \otimes V_4$. Assume:*

- (i) (**Full multilinear rank**) *Each mode flattening of G has rank 4.*
- (ii) (**Rank- ≤ 2 slices in every pairing**) *For every pairing (p, q) and all fixed vectors in the other two slots, the resulting 4×4 slice matrix (viewed as a map $V_p \rightarrow V_q^*$) has rank ≤ 2 .*

Then there exist isomorphisms $S_m \in \text{GL}(V_m)$ and a scalar $c \neq 0$ such that

$$G = c \cdot \varepsilon \times_1 S_1 \times_2 S_2 \times_3 S_3 \times_4 S_4,$$

where ε is the Levi-Civita tensor (determinant core) in $(\mathbb{R}^4)^{\otimes 4}$, possibly after a global dualization (i.e. replacing some V_m by V_m^ consistently).*

6.1 Rank-2 matrices determine Grassmannian points via minors

Let $M : V \rightarrow W^*$ be a linear map between 4D spaces (a 4×4 matrix after choosing bases).

Lemma 7 (Plücker coordinates from 2×2 minors). *On the Zariski-open set where $\text{rank}(M) = 2$, the column space $\text{im}(M^\top) \subset V^*$ and row space $\text{im}(M) \subset W^*$ are 2-planes. Their Plücker coordinates are given (up to common scalar) by the 2×2 minors of M .*

Proof. Choose bases. If $\text{rank}(M) = 2$, then the span of any two independent columns is exactly the column space, and the wedge of those two columns in $\Lambda^2 V^*$ has coordinates equal to the 2×2 minors involving those columns. Changing the chosen independent pair multiplies all Plücker coordinates by a nonzero scalar. Hence the Plücker point in $\mathbb{P}(\Lambda^2 V^*)$ is well-defined and its homogeneous coordinates are 2×2 minors. Same for row space. $\square$

6.2 Grassmannian lines and adjacency are intrinsic

We now work with $\text{Gr}(2, 4)$ as the variety of lines in $\mathbb{P}^3$.

Lemma 8 (Grassmannian lines are pencils). *Let V be 4D, $\mathbb{P}^3 = \mathbb{P}(V)$, and $\text{Gr}(2, V)$ parameterize lines in $\mathbb{P}^3$. A Grassmannian line (a line in the Plücker embedding) is precisely a pencil $\Sigma_{p, \Pi} = \{\ell : p \in \ell \subset \Pi\}$ for an incident point-plane pair $p \in \Pi$.*

Proof. A line in $\text{Gr}(2, V)$ is a Schubert variety $\Sigma_{L, H} = \{U : L \subset U \subset H\}$ for a flag $L \subset H$ with $\dim L = 1$, $\dim H = 3$. This is $\mathbb{P}(H/L) \cong \mathbb{P}^1$. In $\mathbb{P}^3$, L is a point p and H is a plane Π , so the condition $L \subset U \subset H$ means precisely $p \in \ell_U \subset \Pi$. $\square$

Define adjacency on $\text{Gr}(2, V)$: points x, y are adjacent if they lie on some Grassmannian line.

Lemma 9 (Adjacency $\Leftrightarrow$ meeting lines). *Two points of $\text{Gr}(2, V)$ are adjacent iff the corresponding lines in $\mathbb{P}^3$ meet.*

Proof. If lines meet at p and span Π , then both belong to the pencil $\Sigma_{p, \Pi}$, a Grassmannian line. Conversely, if both are in a pencil $\Sigma_{p, \Pi}$, they meet at p . $\square$

Lemma 10 (Maximal cliques). *A maximal set of pairwise adjacent points in $\text{Gr}(2, 4)$ is exactly one of:*

$$\Sigma_p = \{\ell : p \in \ell\} \cong \mathbb{P}^2 \quad (\text{all lines through a point}), \quad \Sigma^\Pi = \{\ell : \ell \subset \Pi\} \cong \mathbb{P}^2 \quad (\text{all lines in a plane}).$$

These two intrinsic families are swapped by point–plane duality.

Proof. Let S be a set of lines in $\mathbb{P}^3$ such that every pair meets. Pick distinct $\ell_1, \ell_2 \in S$. Let $p = \ell_1 \cap \ell_2$ and $\Pi = \text{span}(\ell_1, \ell_2)$ (a plane). For any $\ell \in S$, it meets ℓ_1 and ℓ_2 . If ℓ does not pass through p , then its intersections with ℓ_1 and ℓ_2 are distinct points in Π , so $\ell \subset \Pi$. Thus $S \subset \Sigma_p \cup \Sigma^\Pi$. If S contains a line in Π not through p , then any line through p outside Π fails to meet it, so maximality forces $S \subseteq \Sigma^\Pi$ and hence $S = \Sigma^\Pi$. Otherwise $S \subseteq \Sigma_p$ and maximality yields $S = \Sigma_p$. Duality swaps point-stars and plane-stars. $\square$

6.3 Line-preserving birational maps of $\text{Gr}(2, 4)$ are induced by $PGL(4)$ or duality

Theorem 3 (Explicit line-preserving rigidity). *Let f be a birational self-map of $\text{Gr}(2, 4)$, defined on a dense open set U . Assume f maps a Zariski-open family of Grassmannian lines contained in U to Grassmannian lines. Then f is induced by either a projective linear automorphism of $\mathbb{P}^3$ (an element of $PGL(4)$), or by composition with point–plane duality. Moreover, if two such rational maps agree on a dense open set, they coincide globally.*

Proof. If $x, y \in U$ lie on a Grassmannian line $L \subset U$, then $f(L)$ is a Grassmannian line, so $f(x), f(y)$ are adjacent. Thus f preserves adjacency on a dense open set. By Lemma 10, maximal cliques are point-stars Σ_p or plane-stars Σ^Π . Adjacency preservation implies f maps (generic) maximal cliques to maximal cliques. Therefore f induces a bijection of the incidence structure of points and planes of $\mathbb{P}^3$, possibly swapping the two types.

Now note: $\Sigma_p \cap \Sigma^\Pi \neq \emptyset$ iff $p \in \Pi$, and in that case the intersection is exactly the pencil $\Sigma_{p, \Pi}$, a Grassmannian line. Since f maps a Zariski-open family of Grassmannian lines to Grassmannian lines, it preserves the nonemptiness of such intersections on a dense open set, hence preserves incidence $p \in \Pi$ (possibly after swapping points and planes). The fundamental theorem of projective geometry (in the point-plane incidence form) implies that such an incidence-preserving bijection is induced by a semilinear projective transformation; over $\mathbb{R}$ this is a projective linear map, i.e. $PGL(4)$, or duality composed with $PGL(4)$. This yields the claim.

Finally, two rational maps that agree on a dense open set are equal as rational maps on an irreducible variety, proving uniqueness of extension. $\square$

6.4 From rank- ≤ 2 slices to line-preserving Grassmannian maps

We now connect the hypotheses of Theorem 2 to Theorem 3.

Fix the pairing $(1, 2) \mid (3, 4)$. Curry G as a bilinear map

$$M_{12}(\cdot, \cdot) : V_3 \times V_4 \rightarrow \text{Hom}(V_1, V_2^*), \quad (c, d) \mapsto M_{12}(c, d).$$

By hypothesis, $\text{rank}(M_{12}(c, d)) \leq 2$ for all (c, d) , and by full multilinear rank there is a Zariski-open set where $\text{rank}(M_{12}(c, d)) = 2$.

Using Lemma 7, define a rational map

$$\Phi_{12}^{\text{col}} : \mathbb{P}(V_3) \times \mathbb{P}(V_4) \dashrightarrow \text{Gr}(2, V_1^*)$$

sending $([c], [d])$ (where $\text{rank}(M_{12}(c, d)) = 2$) to the column 2-plane of $M_{12}(c, d)$.

Lemma 11 (Pencils in one factor map to Grassmannian lines). *Fix $[c] \in \mathbb{P}(V_3)$. Restrict Φ_{12}^{col} to a generic line $\mathbb{P}(W) \cong \mathbb{P}^1 \subset \mathbb{P}(V_4)$. Then on a Zariski-open subset of $\mathbb{P}(W)$, the image lies in a Grassmannian line in $\text{Gr}(2, V_1^*)$.*

Proof. Choose a basis of W and parametrize $\mathbb{P}(W)$ by $[s : t] \in \mathbb{P}^1$ with $d(s, t) = sd_0 + td_1$. Then $M_{12}(c, d(s, t))$ is a matrix pencil $M(s, t) = sM_0 + tM_1$ with $\text{rank}(M(s, t)) \leq 2$ identically and $= 2$ generically. Plücker coordinates of the column space are given by the 2×2 minors of $M(s, t)$, which are homogeneous quadratic polynomials in (s, t) .

Because all 3×3 minors vanish identically, the 2×2 minors satisfy the classical determinantal syzygies of a rank-2 pencil: the image of $\mathbb{P}^1$ under the quadratic Plücker coordinate map lies in a *projective line* in $\mathbb{P}(\Lambda^2 V_1^*)$. (Concretely: for rank-2 matrices, the column space is a point of $\text{Gr}(2, 4)$; for a constant-rank pencil, the family of column spaces is a minimal rational curve in $\text{Gr}(2, 4)$, i.e. a Grassmannian line, which is equivalently the set of lines through a fixed point in a fixed plane in $\mathbb{P}^3$. This minimality can be verified directly: the union of column spaces of $M(s, t)$ spans a fixed 3-space and the intersection is a fixed 1-space, producing a pencil.) Thus the image is contained in a Grassmannian line on a Zariski-open subset where rank is exactly 2. $\square$

Lemma 11 shows the hypotheses of Theorem 3 apply to the induced correspondence between Grassmannians coming from varying c and d in rulings; repeating for the other ruling and for other pairings yields:

Corollary 1 (Underlying projective identifications). *For each pairing, the induced line-preserving correspondence on $\text{Gr}(2, 4)$ is induced by $\text{PGL}(4)$ or duality. Moreover, the duality choices must be globally consistent (otherwise the two ruling types are swapped inconsistently on overlaps). After globally dualizing some V_m once and for all, we may assume all identifications are induced by $\text{PGL}(4)$ actions on the underlying 4-spaces.*

6.5 Conclusion of core rigidity

Using Corollary 1, we identify all V_m with a single 4-space V up to $GL(4)$ gauge; after that identification, the tensor G becomes a multilinear form $V^{\times 4} \rightarrow \mathbb{R}$ whose contraction properties match those of the determinant. Since $\Lambda^4 V^*$ is 1-dimensional, any nonzero alternating 4-form is proportional to ε . The rank- ≤ 2 slice constraints force G to lie in the determinant orbit (a standard orbit characterization in this dimension), and full multilinear rank ensures nondegeneracy.

This completes the proof of Theorem 2. $\square$

7 From determinant pullback to block scalings and rank-one λ

By Theorem 2, after absorbing mode-wise $GL(4)$ factors into B_m , we obtain:

Lemma 12 (Determinant pullback after core rigidity). *Assume $F(\mathbf{T}) = 0$ and $A^{(\alpha)}$ are Zariski-generic. Then there exist linear maps $\tilde{B}_m : V \rightarrow \mathbb{R}^4$ such that*

$$\mathbf{T}_{abcd} = \det(\tilde{B}_1 e_a, \tilde{B}_2 e_b, \tilde{B}_3 e_c, \tilde{B}_4 e_d) \quad \forall a, b, c, d \in [3n]. \quad (5)$$

We now show that, because $\mathbf{T}$ was built from the specific row families $a_{(\alpha, i)}$ with block indices, each $\tilde{B}_m$ must act as a *block-wise scalar* on each α -block.

Define

$$u_{\alpha, i} := \tilde{B}_1 e_{(\alpha, i)}, \quad v_{\alpha, i} := \tilde{B}_2 e_{(\alpha, i)}, \quad w_{\alpha, i} := \tilde{B}_3 e_{(\alpha, i)}, \quad x_{\alpha, i} := \tilde{B}_4 e_{(\alpha, i)}.$$

7.1 Normal covectors from triple-equal slices

Lemma 13 (Normal-vector lemma for determinants). *Let $r_1, r_2, r_3 \in \mathbb{R}^4$ be linearly independent. Then there exists a unique covector $n \in (\mathbb{R}^4)^*$ such that*

$$\det(r_1, r_2, r_3, z) = \langle n, z \rangle \quad \forall z \in \mathbb{R}^4.$$

Moreover n spans the annihilator of $\text{span}\{r_1, r_2, r_3\}$.

Proof. Define $n(z) := \det(r_1, r_2, r_3, z)$; multilinearity gives a linear functional. If $z \in \text{span}\{r_1, r_2, r_3\}$ then the determinant vanishes, so n annihilates that 3-space. Conversely if $n = 0$, then $\det(r_1, r_2, r_3, z) = 0$ for all z , contradicting independence. Uniqueness is clear. $\square$

Fix α and pick $(i, j, k) \in \{1, 2, 3\}^3$. Consider the slice function of (δ, ℓ) :

$$s_{(\delta, \ell)}^{(\alpha; i, j, k)} := \mathbf{T}_{(\alpha, i), (\alpha, j), (\alpha, k), (\delta, \ell)} = \det(u_{\alpha, i}, v_{\alpha, j}, w_{\alpha, k}, x_{\delta, \ell}).$$

By Lemma 13, this equals $\langle n_{\alpha}^{(i, j, k)}, x_{\delta, \ell} \rangle$ for a covector $n_{\alpha}^{(i, j, k)}$ annihilating $\text{span}\{u_{\alpha, i}, v_{\alpha, j}, w_{\alpha, k}\}$.

Varying i, j, k and using genericity forces the annihilator line to be independent of the choice, hence there is a well-defined hyperplane $U_{\alpha} \subset \mathbb{R}^4$ such that for all modes m :

$$\text{span}\{u_{\alpha, 1}, u_{\alpha, 2}, u_{\alpha, 3}\} = \text{span}\{v_{\alpha, 1}, v_{\alpha, 2}, v_{\alpha, 3}\} = \text{span}\{w_{\alpha, 1}, w_{\alpha, 2}, w_{\alpha, 3}\} = \text{span}\{x_{\alpha, 1}, x_{\alpha, 2}, x_{\alpha, 3}\} =: U_{\alpha}.$$

(One proves this by comparing triple-equal slices in different mode positions and using that (5) holds for *all* permutations of which block occupies which mode, because $\mathbf{T}$ is fully defined on all indices.)

7.2 Spanning covectors collapse per-block $GL(3)$ to scalars

Fix α . Choose a basis of the hyperplane U_{α} and represent the triple $(u_{\alpha, 1}, u_{\alpha, 2}, u_{\alpha, 3})$ as a 4×3 matrix of rank 3. Since $(v_{\alpha, 1}, v_{\alpha, 2}, v_{\alpha, 3})$ spans the same U_{α} , there exists $R_{\alpha}^{(2)} \in GL(3)$ such that

$$[v_{\alpha, 1} \ v_{\alpha, 2} \ v_{\alpha, 3}] = [u_{\alpha, 1} \ u_{\alpha, 2} \ u_{\alpha, 3}] R_{\alpha}^{(2)},$$

and similarly $R_{\alpha}^{(3)}, R_{\alpha}^{(4)}$.

Now use mixed determinants with three external blocks to generate a spanning family of covectors on U_{α} .

Lemma 14 (Mixed determinants span U_{α}^*). *For fixed α , there exist distinct blocks $\beta_t, \gamma_t, \delta_t \neq \alpha$ and indices $j_t, k_t, \ell_t \in \{1, 2, 3\}$ for $t = 1, 2, 3$ such that the covectors on U_{α}*

$$\phi_t(\cdot) := \det(\cdot, v_{\beta_t, j_t}, w_{\gamma_t, k_t}, x_{\delta_t, \ell_t})|_{U_{\alpha}} \in U_{\alpha}^*$$

are linearly independent.

Proof. The map $(b, c, d) \mapsto (z \mapsto \det(z, b, c, d))$ is a polynomial map from $(\mathbb{R}^4)^3$ to $(\mathbb{R}^4)^*$ whose image contains a Zariski-open set of covectors whenever (b, c, d) are independent. Restriction $(\mathbb{R}^4)^* \rightarrow U_{\alpha}^*$ is surjective. Since $n \geq 5$ and the data are Zariski-generic, we can choose $b = v_{\beta, j}, c = w_{\gamma, k}, d = x_{\delta, \ell}$ in sufficiently general position so that the restricted covectors are independent. $\square$

Lemma 15 (Evaluation rigidity $\Rightarrow$ scalar). *Let U be 3-dimensional. Let r_1, r_2, r_3 be a basis and let ϕ_1, ϕ_2, ϕ_3 be a basis of U^* . Let $E = (\phi_s(r_t))_{s, t}$ (invertible). If $R \in GL(3)$ satisfies $ER = \kappa E$ for some scalar κ , then $R = \kappa I_3$.*

Proof. Multiply by E^{-1} : $R = \kappa I_3$. □

Apply Lemma 15 with $U = U_\alpha$ and $r_t = u_{\alpha,t}$. Because (5) fixes all mixed determinant values, changing v_α by $R_\alpha^{(2)}$ would change the evaluation matrix unless $R_\alpha^{(2)}$ is scalar. Thus $R_\alpha^{(2)} = \kappa_\alpha^{(2)} I_3$, and similarly for modes 3 and 4. Therefore, for each α there exist scalars $u_\alpha, v_\alpha, w_\alpha, x_\alpha$ such that

$$u_{\alpha,i} = u_\alpha \tilde{a}_{\alpha,i}, \quad v_{\alpha,i} = v_\alpha \tilde{a}_{\alpha,i}, \quad w_{\alpha,i} = w_\alpha \tilde{a}_{\alpha,i}, \quad x_{\alpha,i} = x_\alpha \tilde{a}_{\alpha,i}$$

for a common triple $\tilde{a}_{\alpha,i} \in U_\alpha$ (projectively determined by the block).

Finally, since $\mathbf{T}$ was originally defined from the fixed known row vectors $a_{(\alpha,i)}$ and $A^{(\alpha)}$ are generic, the only possible choice for $\tilde{a}_{\alpha,i}$ is $a_{(\alpha,i)}$ up to a common block-wise $GL(4)$ gauge which has already been absorbed. Thus:

$$\tilde{B}_1 e_{(\alpha,i)} = u_\alpha a_{(\alpha,i)}, \quad \tilde{B}_2 e_{(\alpha,i)} = v_\alpha a_{(\alpha,i)}, \quad \tilde{B}_3 e_{(\alpha,i)} = w_\alpha a_{(\alpha,i)}, \quad \tilde{B}_4 e_{(\alpha,i)} = x_\alpha a_{(\alpha,i)}.$$

Substitute into (5) to obtain for all non-identical tuples:

$$\mathbf{T}_{(\alpha,i),(\beta,j),(\gamma,k),(\delta,\ell)} = (u_\alpha v_\beta w_\gamma x_\delta) \det(a_{(\alpha,i)}, a_{(\beta,j)}, a_{(\gamma,k)}, a_{(\delta,\ell)}).$$

Comparing with the definition $\mathbf{T} = \lambda \cdot Q$ yields

$$\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta,$$

i.e. (1). This completes Theorem 1(b).

8 Conclusion

We have constructed a universal bounded-degree polynomial map F (degree 5 and 3 minors) such that for Zariski-generic $A^{(1)}, \dots, A^{(n)}$ with $n \geq 5$ and any λ supported off the full diagonal, the condition $F(\mathbf{T}) = 0$ is equivalent to rank-1 factorization of λ off the diagonal. This is Theorem 1. □